File

Name: P2-K01-geometry-mapped
Description: KISSsoft example
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1 Overview

Calculation method		ISO 6336:2019	
Drawing or article number:			
Gear 1:		0.000.0	
Gear 2:		0.000.0	
		------- Gear 1	---------- Gear 2 -------
Power (kW)	[P]	75.000	
Speed (1/min)	[n]	440.8	149.1
Number of load cycles (in mio.)	[NL]	528.960	178.913
Torque (Nm)	[T]	1624.8	4803.7
Application factor	[KA]	1.25	
Required service life (h)	[H]	20000.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:		Right flank	
Gear 1 direction of rotation:		Clockwise	

2 Tooth geometry

Geometry calculation according to		ISO 21771:2007 (withdrawn)	
		------- Gear 1	---------- Gear 2 -------
Center distance (mm)	[a]	274.056	
Center distance tolerance		ISO 286:2010 Measure js6	
Normal module (mm)	[mn]	6.0000	
Transverse module (mm)	[mt]	6.0000	
Normal diametral pitch (1/in)	[Pnd]	4.23333	
Transverse diametral pitch (1/in)	[Ptd]	4.23333	
Normal pressure angle (°)	[αn]	20.0000	
Helix angle at reference circle (°)	[β]	0.0000	
Number of teeth	[z]	23	68
Facewidth (mm)	[b]	28.30	28.30
Hand of gear		Spur gear	
Accuracy grade	[Q-ISO 1328:2013]	A6	A6
Inner diameter (mm)	[di]	67.73	329.50
Shaft diameter (mm)	[ds]	77.58	377.50
Inner diameter of gear rim (mm)	[dbi]	0.00	0.00
		------- Gear 1	---------- Gear 2 -------

3 Materials

Gear 1

18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

Gear 2

18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

		------- Gear 1	---------- Gear 2 -------
Surface hardness		HRC 61	HRC 61
Material treatment according to ISO 6336:2006 Normal, Life factors Z_{NT} and Y_{NT} ≥ 0.85			
Infinite life strength for tooth root stress (N/mm²)	[σFlim]	425.00	425.00
Infinite life strength for Hertzian pressure (N/mm²)	[σHlim]	1500.00	1500.00
Young's modulus (N/mm²)	[E]	206000	206000
Poisson's ratio	[v]	0.300	0.300
Tensile strength (N/mm²)	[σB]	1200.00	1200.00
Yield point (N/mm²)	[σS]	850.00	850.00

3.1 Gear roughness

		------- Gear 1	---------- Gear 2 -------
Arithmetic mean roughness value R_a, flank (µm)	[RAH]	0.60	0.60
Arithmetic mean roughness value R_a, root (µm)	[RAF]	3.00	3.00
Mean peak-to-valley roughness R_z, flank (µm)	[RZH]	4.80	4.80
Mean peak-to-valley roughness R_z, root (µm)	[RZF]	20.00	20.00

3.2 Lubrication

Lubrication type	Oil bath lubrication	
Type of oil	Klüberoil GEM 1-220 N with details about wear coefficient kw	
Lubricant base	Mineral-oil base	
Lubricant factor	[XL]	1.000
Oil nominal kinematic viscosity at 40°C (mm²/s)	[v40]	220.00
Oil nominal kinematic viscosity at 100°C (mm²/s)	[v100]	19.00
Specific density at 15°C (kg/dm³)	[ρ]	0.890
Oil temperature (°C)	[TS]	80.000

4 Geometry

4.1 Reference profiles

Reference profile Gear 1

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.472
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	71.562
(mm)	[a0.e/i]	70.360 /70.292
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	2.388 / 2.394

Reference profile Gear 2

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.472
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000

Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	202.509
(mm)	[a0.e/i]	202.330 /202.248
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	2.505 / 2.510

4.1.1 Information on final machining

		------- Gear 1	---------- Gear 2 -------
Dedendum reference profile	[hfP*]	1.250	1.250
Tooth root radius reference profile	[ρfP*]	0.380	0.380
Addendum reference profile	[haP*]	1.000	1.000
Protuberance height coefficient	[hprP*]	0.000	0.000
Protuberance angle (°)	[αprP]	0.000	0.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000
Type of profile modification:	none (only running-in)		
Tip relief by running in (µm)	[Ca L/R]	2.0 / 2.0	2.0 / 2.0

4.2 Basic data

Overall transmission ratio	[itot]	-2.957
Gear ratio	[u]	2.957
Transverse module (mm)	[mt]	6.000
Transverse pressure angle (°)	[αt]	20.000
Working pressure angle (°)	[αwt]	20.598
(°)	[αwt.e/i]	20.607 /20.589
Working pressure angle at normal section (°)	[αwn]	20.598
Base helix angle (°)	[βb]	0.000
Helix angle at operating pitch circle (°)	[βw]	0.000
Reference center distance (mm)	[ad]	273.000
Sum of profile shift coefficients	[Σxi]	0.1785

		------- Gear 1	---------- Gear 2 -------
Profile shift coefficient	[x]	0.4270	-0.2485
Generating profile shift coefficient	[xE.e/i]	0.4052/ 0.3938	-0.2783/-0.2920
Virtual gear no. of teeth	[zn]	23.000	68.000
Involute length (mm)	[l_dFa-l_dFf]	12.538	11.555

4.3 Diameters and their allowances

		------- Gear 1	---------- Gear 2 -------
Reference diameter (mm)	[d]	138.000	408.000
Base diameter (mm)	[db]	129.678	383.395
Tip alteration (mm)	[k*mn]	-0.015	-0.015
Tip diameter (mm)	[da]	155.094	416.988
(mm)	[da.e/i]	155.094 /154.994	416.988 /416.888
Tip diameter allowances (mm)	[Ada.e/i]	0.000 / -0.100	0.000 / -0.100
Tip form diameter (mm)	[dFa]	155.094	416.988
(mm)	[dFa.e/i]	155.094 /154.994	416.988 /416.888
Active tip diameter (mm)	[dNa]	155.094	416.988
(mm)	[dNa.e/i]	155.094 /154.994	416.988 /416.888
V-Circle diameter (mm)	[dv]	143.124	405.018
(mm)	[dv.e/i]	142.863 /142.726	404.661 /404.496

Operating pitch diameter (mm)	[dw]	138.534	409.578
(mm)	[dw.e/i]	138.542 /138.526	409.602 /409.554
Root diameter (mm)	[df]	128.124	390.018
(mm)	[df.e/i]	127.863 /127.726	389.661 /389.496
Active root diameter (mm)	[dNf]	132.849	398.249
(mm)	[dNf.e/i]	132.925 /132.829	398.323 /398.224
Root form diameter (mm)	[dFf]	132.478	395.168
(mm)	[dFf.e/i]	132.324 /132.245	394.916 /394.801

4.4 Tip clearances and tooth heights

		------- Gear 1 ----------	Gear 2 -------
Theoretical tip clearance (mm)	[c]	1.500	1.500
Effective tip clearance (mm)	[c.e/i]	1.827 / 1.662	1.765 / 1.614
Reserve (dNf-dFf)/2 (mm)	[cF.e/i]	0.340 / 0.253	1.761 / 1.654
Addendum, $m_n(h_{aP}$*+x+k) (mm)	[ha]	8.547	4.494
(mm)	[ha.e/i]	8.547 / 8.497	4.494 / 4.444
Dedendum, $m_n(h_{fP}$*-x) (mm)	[hf]	4.938	8.991
(mm)	[hf.e/i]	5.069 / 5.137	9.170 / 9.252
Tooth height (mm)	[h]	13.485	13.485

4.5 Roll angle

		------- Gear 1 ----------	Gear 2 -------
Roll angle at dFa (°)	[ξFa.e/i]	37.589 /37.509	24.505 /24.467
Roll angle to dNf (°)	[ξNf.e/i]	12.902 /12.709	16.144 /16.090
Roll angle at dFf (°)	[ξFf.e/i]	11.635 /11.457	14.152 /14.080

4.6 Tooth thickness and pitch

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness on reference circle, arc, in module	[sn*]	1.8816	1.3899
Normal tooth thickness at tip circle (mm)	[san]	3.300	4.921
(mm)	[san.e/i]	3.257 / 3.137	4.830 / 4.727
Normal tooth thickness at tip form circle (mm)	[sFan]	3.300	4.921
(mm)	[sFan.e/i]	3.257 / 3.137	4.830 / 4.727
Normal space width at root circle (mm)	[efn]	0.000	5.062
(mm)	[efn.e/i]	0.000 / 0.000	5.116 / 5.142
Pitch on reference circle (mm)	[pt]	18.850	18.850
Base pitch (mm)	[pbt]	17.713	17.713
Transverse pitch on contact-path (mm)	[pet]	17.713	17.713

4.7 Sliding

		------- Gear 1 ----------	Gear 2 -------
Max. sliding velocity at tip (m/s)	[vga]	1.122	0.614
Specific sliding at the tip	[ζa]	0.572	0.480
Specific sliding at the root	[ζf]	-0.922	-1.334
Mean specific sliding	[ζm]	0.539	

Sliding factor on tip	[Kga]	0.351	0.192
Sliding factor on root	[Kgf]	-0.192	-0.351

4.8 Contact ratios

		Pair
Minimal length of contact line (mm)	[Lmin]	28.300
Transverse contact ratio	[εα]	1.587
	[εα.e/m/i]	1.590 / 1.581/ 1.572
Overlap ratio	[εβ]	0.000
Total contact ratio	[εγ]	1.587
	[εγ.e/m/i]	1.590 / 1.581/ 1.572
Length of path of contact (mm)	[gα]	28.110
(mm)	[gα.e/m/i]	28.156 /28.001/27.846
Approach path of contact (mm)	[gf]	9.941
(mm)	[gf.e/m/i]	9.975 / 9.877/ 9.780
Recess path of contact (mm)	[ga]	18.169
(mm)	[ga.e/m/i]	18.181 /18.124/18.067

		Gear 1	Gear 2
Addendum contact ratio	[ε]	1.026	0.561
	[ε.e/i]	1.026 / 1.020	0.563 / 0.552
Length T1-A and T2-A (mm)	[T1A,T2A]	14.428	81.987
(mm)	[.e/i]	14.382 / 14.600	81.987 / 81.860
Length T1-B and T2-B (mm)	[T1B,T2B]	24.825	71.590
(mm)	[.e/i]	24.825 / 24.734	71.544 / 71.726
Length T1-C and T2-C (mm)	[T1C,T2C]	24.369	72.046
(mm)	[.e/i]	24.357 / 24.380	72.012 / 72.080
Length T1-D and T2-D (mm)	[T1D,T2D]	32.140	64.274
(mm)	[.e/i]	32.095 / 32.313	64.274 / 64.147
Length T1-E and T2-E (mm)	[T1E,T2E]	42.538	53.877
(mm)	[.e/i]	42.538 / 42.447	53.831 / 54.014
Length T1-T2 (mm)	[T1T2]	96.415	
(mm)	[.e/i]	96.369 / 96.460	
Diameter of single contact point B (mm)	[d-B]	138.858	409.258
(mm)	[d-B.e/i]	138.858 /138.792	409.226 /409.353
Diameter of single contact point D (mm)	[d-D]	144.735	404.371
(mm)	[d-D.e/i]	144.695 /144.889	404.371 /404.291

5 General influence factors

5.1 Forces and circumferential speed

		Gear 1	Gear 2
Nominal circum. force at pitch circle (N)	[Ft]	23547.35	
Axial force (N)	[Fa]	0.00	0.00
Radial force (N)	[Fr]	8570.53	8570.53
Normal force (N)	[Fnorm]	25058.56	25058.56
Nominal circumferential force per mm (N/mm)	[w]	832.06	
Only as information: Forces at operating pitch circle:			
Nominal circumferential force (N)	[Ftw]	23456.63	
Axial force (N)	[Faw]	0.00	0.00
Radial force (N)	[Frw]	8815.78	

Circumferential speed reference circle (m/s)	[v]	3.19	
Circumferential speed operating pitch circle (m/s)	[v(dw)]	3.20	

5.2 Contact stiffness

Running-in value (µm)	[yp]	0.8	
Running-in value (µm)	[yf]	0.9	
Correction factor	[CM]	0.800	
Gear blank factor	[CR, bs/b, sr/mn]	0.898	(0.250,5.000)
Basic rack factor	[CB]	0.975	
Material coefficient	[E/Est]	1.000	
Single stiffness (N/mm/µm)	[c']	12.655	
Theoretical single stiffness (N/mm/µm)	[c'th]	18.067	
Contact stiffness (N/mm/µm)	[cγα]	18.226	
Contact stiffness (N/mm/µm)	[cγβ]	15.492	
Reduced mass (kg/mm)	[mRed]	0.05682	
Resonance speed (min-1)	[nE1]	7436	
Resonance ratio (-)	[N]	0.059	
Subcritical range			

5.3 Calculation of K factors

Running-in value (µm)	[yα]	0.9	
Bearing distance l of pinion shaft (mm)	[l]	56.600	
Distance s of pinion shaft (mm)	[s]	5.660	
Outside diameter of pinion shaft (mm)	[dsh]	28.300	
Load in accordance with Figure 13, ISO 6336-1:2006	[-]	4	
0:a), 1:b), 2:c), 3:d), 4:e)			
Coefficient K' according to Figure 13, ISO 6336-1:2006	[K']	-0.60	
With stiffening			
Effective flank line deviation (µm)	[Fβy]	4.50	
from deformation of shaft (µm)	[fsh·B1]	5.44	
fsh (µm)	[fsh]	5.44	
B1	[B1]	1.00	
fHβ5 (µm)	[fHβ5]	7.00	
Tooth without flank line modification			
Position of contact pattern:	favorable		
from production tolerances (µm)	[fmα*B2]	13.09	
B_2=1.00			
Tooth trace deviation, theoretical (µm)	[Fβx]	5.30	
Running-in value (µm)	[yβ]	0.79	

5.4 K factors

Dynamic factor	[Kv]	1.019
Face load factors		
- Flank	[KHβ]	1.033
- Tooth root	[KFβ]	1.023
- Scuffing	[KBβ]	1.033
Transverse load factors		
- Flank	[KHα]	1.000
- Tooth root	[KFα]	1.000
- Scuffing	[KBα]	1.000
Application factor	[KA]	1.250
Mesh load factor	[Kγ]	1.000

6 Calculation of tooth root strength (fracture)

Calculation of Tooth form coefficients according method: B

		------- Gear 1	---------- Gear 2 -------
Calculated with generating profile shift coefficient	[xE.i]	0.3938	-0.2920
Tooth form factor	[YF]	1.208	1.588
Stress correction factor	[YS]	2.220	1.843
Load application angle (°)	[αFen]	22.920	19.399
Load distribution influence factor	[fε]	1.000	
Load application diameter (mm)	[d_{en}]	144.735	409.258
Bending moment arm (mm)	[hFe]	5.858	7.281
(-)	[hFe*]	0.976	1.213
Tooth root thickness at critical cross section (mm)	[sFn]	13.081	12.870
(-)	[sFn*]	2.180	2.145
Root fillet radius at critical cross section (mm)	[ρF]	2.678	3.400
(-)	[ρF*]	0.446	0.567
Tangent contact point (mm)	[x,y]	6.541 /64.539	
Tangent contact point (mm)	[x,y]		6.435 /195.955
Diameter at critical cross section (mm)	[d_{sFn}]	129.739	392.120
Tangent at critical cross section (°)	[α_{sFn}]	30.000	30.000
Notch parameter	[q_s]	2.443	1.893
Helix angle factor	[Yβ]	1.000	
Deep tooth factor	[YDT]	1.000	
Gear rim factor	[YB]	1.00	1.00
Effective facewidth (mm)	[beff]	28.30	28.30
Nominal stress at tooth root (N/mm²)	[σF0]	371.93	406.04
Tooth root stress (N/mm²)	[σF]	484.29	528.70
Permissible bending stress at root of Test-gear			
Notch sensitivity factor	[YδrelT]	0.999	0.994
Surface factor	[YRrelT]	0.957	0.957
Size factor, tooth root	[YX]	0.990	0.990
Life factor	[YNT]	0.902	0.921
$Y_{\delta relT} \cdot Y_{RrelT} \cdot Y_X \cdot Y_{NT}$		0.853	0.867
Alternating bending factor, mean stress influence coefficient			
	[YM]	1.000	1.000
Stress correction factor	[YST]	2.00	
$Y_{ST} \cdot \sigma_{Flim}$ (N/mm²)	[σFE]	850.00	850.00
Permissible tooth root stress σFG/SFmin (N/mm²)	[σFP]	518.18	526.63
Limit strength tooth root (N/mm²)	[σFG]	725.45	737.29

6.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SFmin]	1.40	1.40
Safety for tooth root stress	[SF=σFG/σF]	1.50	1.39
Transmittable power (kW)	[kWRating]	80.25	74.71

7 Calculation of flank strength (pitting)

		------- Gear 1	---------- Gear 2 -------
Zone factor	[ZH]	2.455	
Elasticity coefficient (√N/mm)	[ZE]	189.812	
Contact ratio factor	[Zε]	0.897	
Helix angle factor	[Zβ]	1.000	
Effective facewidth (mm)	[beff]	28.30	
Nominal contact stress (N/mm²)	[σH0]	1187.07	
Contact stress at operating pitch circle (N/mm²)	[σHw]	1361.32	

Single tooth contact factor	[ZB,ZD]	1.00	1.00
Contact stress (N/mm²)	[σHB, σHD]	1361.32	1361.32
Lubricant coefficient for N_L	[ZL]	1.020	1.020
Speed factor at N_L	[ZV]	0.973	0.973
Roughness factor for N_L	[ZR]	0.979	0.979
Material hardening factor for N_L	[ZW]	1.000	1.000
Life factor	[ZNT]	0.930	0.962
Size factor (flank)	[ZX]	1.000	1.000
$Z_L \cdot Z_V \cdot Z_R \cdot Z_{NT} \cdot Z_X$		0.903	0.934
Limited pitting is permissible: No			
Permissible contact stress, σHG/SHmin (N/mm²)	[σHP]	1354.47	1400.27
Pitting stress limit (N/mm²)	[σHG]	1354.47	1400.27

7.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SHmin]	1.00	1.00
Safety factor for contact stress on operating pitch circle	[SHw]	0.99	1.03
Safety against pressure, σHG/σHBD Single contact	[SHBD]	0.99	1.03
Safety regarding transmittable torque	[SHBD²]	0.99	1.06
Transmittable power (kW)	[kWRating]	74.25	79.35

8 Micropitting

Calculation method according to ISO/TS 6336-22:2018

Lubricant load according to FVA Info sheet 54/7 10, Klüberoil GEM 1-220 N with details about wear coefficient kw

Reference data FZG-C Test:

Torque (Nm)	[T1Ref]	265.100
Line load at contact point A (N/mm)	[FbbRef,A]	236.300
Oil temperature (°C)	[θOilRef]	90.000
Tooth mass temperature (°C)	[θMRef]	127.666
Contact temperature (°C)	[θBRef,A]	251.735
Lubrication gap thickness (µm)	[hRef,A]	0.051
Specific film thickness in test	[λGFT]	0.102

		------- Gear 1	---------- Gear 2 -------
Calculation of permissible specific film thickness			
Material coefficient	[WW]	1.000	
Permissible specific film thickness	[λGFP]	0.143	
Interim results in accordance with ISO/TS 6336-22:2018			
Coefficient of friction	[μ_m]	0.069	
Lubricant factor	[XL]	1.000	
Roughness factor	[XR]	0.937	
Lubrication coefficient for lubrication type	[XS]	1.000	
Tooth mass temperature (°C)	[θM]	87.884	
Tip relief factor	[XCa(A)]	1.000	
Loss factor	[HV]	0.143	
Equivalent Young's modulus (N/mm²)	[Er]	226373.626	
Pressure-viscosity coefficient (m²/N)	[α38]	0.02162	
Dynamic viscosity (Ns/m²)	[ηtM]	22.926	
Roughness average value (µm)	[Ra]	0.600	

Calculation of speeds, load distribution and flank curvature according to method B following ISO/TS 6336-22:2018.

C_a taken as optimal in the calculation. 0=no, 1=yes	0	0
Calculation at point (0:A, 1:AB, 2:B, 3:C, 4:D, 5:DE, 6:E, -1:No Point)	0	

Diameter (mm)	[dy]	132.849	416.988
Relative radius of curvature (mm)	[ρred]	12.269	
Load sharing factor	[XY]	0.333	
Contact stress (N/mm²)	[pH]	930.997	
Contact stress (N/mm²)	[pdyn]	1067.655	
Minimal specific film thickness	[λGFY]	0.196	(h_Y=0.118 µm)
Safety against micropitting	[SΛ(B)]	1.373	

9 Safety of hardened layer (subsurface fatigue)

CHD according to		DNV-CG-0036:2021	
		------- Gear 1 ----------	Gear 2 -------
Required hardening depth CHD (mm)	[t550]	0.50 / 0.60	0.50 / 0.60
Required hardness at t550 (HV)	[HV.t]	550	550
Required hardening depth CHD (mm)	[t400]	0.90 / 1.10	0.90 / 1.10
Required hardness at t400 (HV)	[HV.t]	400	400
Required hardening depth CHD (mm)	[t300]	1.30 / 1.60	1.30 / 1.60
Required hardness at t300 (HV)	[HV.t]	300	300
Condition for hardness on the surface (HV)	[> HV.0]	600	600
Condition for hardness on the surface (HRC)	[> HRC.0]	55	55

HV.0 is calculated with t_z/a_H = 0.5, S_H = min(S_{EHT}, S_{Hssmin})

Safety of the hardened layer	[SEHT]	0.99	0.99
Required safety	[SHssmin]	1.00	1.00
Result		2	2

-1:Not checked, 0:CHD too shallow, 1:CHD too deep or HV.0 too large, 2:Check performed,
3:Check performed with insufficient hardening depths (t_z/a_H<0.5, t_z/a_{Hst}<0.6)

10 Tooth flank fracture

Calculation method according to ISO/TS 6336-4 (2019)

Calculation according to Method B

		------- Gear 1 ----------	Gear 2 -------
Calculation at point (0:A, 1:AB, 2:B, 3:C, 4:D, 5:DE, 6:E, -1:No Point)		2	2
Diameter (mm)	[dy]	138.858	409.258
Relative radius of curvature (mm)	[ρred]	18.433	18.433
Load sharing factor	[XCP]	1.000	1.000
Contact stress (N/mm²)	[pH]	1315.555	1315.555
Contact stress (N/mm²)	[pdyn]	1508.661	1508.661
Required hardening depth CHD (mm)	[t550]	0.50 / 0.60	
Required hardening depth CHD (mm)	[t550]		0.50 / 0.60
The calculation will be performed with the minimum hardening depth			
Core hardness (HV)	[HV_{core}]	342	342
Hardness curve from:			
HV(y) gear 1		ISO/TS 6336-4, chapter 7.3.3, Method C1	
HV(y) gear 2		ISO/TS 6336-4, chapter 7.3.3, Method C1	
Maximum material exposure	[AFFmax]	1.278	1.277
Depth of maximum load (mm)	[y]	-1.050	-1.050
Material coefficient	[$K_{material}$]	1.131	1.131
Material shearing strength (N/mm²)	[τper]	164.199	164.250
Equivalent stress state (N/mm²)	[τeff]	203.239	203.239
Safety factor according to ISO DTS 19042-1	[SFF]	0.877	0.877

(S_{FF} = 1/($A_{FF,max}$ + 0.2) + 0.2)

11 Scuffing load capacity

Calculation method according to ISO/TS 6336-20/21:2022

		------- Gear 1	---------- Gear 2 -------
Thermal contact coefficient (N/mm/s^.5/K)	[BM]	13.780	13.780
Relevant tip relief (µm)	[Ca]	2.00	2.00
Optimal tip relief (µm)	[Ceff]	82.19	
C_a taken as optimal in the calculation. 0=no, 1=yes		0	0

		---------- Pair ----------	
Helical load factor for scuffing	[KBγ]	1.000	
Lubrication coefficient for lubrication type	[XS]	1.000	
Scuffing test, load stage	[FZGtest]	FZG - Test A / 8.3 / 90 (ISO 14635 - 1),	14
Multiple meshing factor	[Xmp]	1.000	
Relative structural factor, scuffing	[XWrelT]	1.000	
Effective facewidth (mm)	[beff]	28.300	
Applicable circumferential force/facewidth (N/mm)	[wBt]	1094.261	
$K_{B\gamma}$ = 1.000, $w_{Bt} \cdot K_{B\gamma}$ = 1094.261			
Angle factor	[Xαβ]	0.986	
ε_1 =1.026, ε_2 =0.561			

11.1 Flash temperature-criteria

		---------- Pair ----------		
Lubricant factor	[XL]	0.844		
Tooth mass temperature (°C)	[θMi]	94.10		
$\theta_{Mi} = \theta_{oil} + X_S \cdot 0.47 \cdot X_{mp} \cdot \theta_{flm}$				
Average flash temperature (Δ°C)	[θflm]	29.99		
Scuffing temperature (°C)	[θS]	451.99		
Γ coordinates (point of highest temperature)	[Γ]	0.652		
[Γ.A]= -0.408, [Γ.E]= 0.746				
Maximum flash temperature (Δ°C)	[θflamax]	48.20		
Maximum contact temperature (°C)	[θBmax]	142.30		
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058		
Approach factor	[XJ]	1.000		
Load sharing factor	[XΓ]	0.406		
Dynamic viscosity (mPa*s)	[ηM]	30.07	(80.0	°C)
Coefficient of friction	[μ_m]	0.076		
Required safety	[SBmin]	2.000		
Margin of safety for scuffing, flash temperature	[SB]	5.971		

11.2 Integral temperature-criteria

		---------- Pair ----------		
Lubricant factor	[XL]	1.000		
Tooth mass temperature (°C)	[θMC]	94.30		
$\theta_{MC} = \theta_{oil} + X_S \cdot 0.70 \cdot \theta_{flaint}$				
Integral scuffing temperature (°C)	[θSint]	455.17		
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058		
Running-in factor, well run in	[XE]	1.000		
Contact ratio factor	[Xε]	0.242		
Dynamic viscosity (mPa*s)	[ηOil]	30.07	(80.0	°C)
Mean coefficient of friction	[μ_m]	0.069		
Geometry factor	[XBE]	0.330		
Meshing factor	[XQ]	1.000		
Tip relief factor	[XCa]	1.105		
Mean flash temperature (Δ°C)	[θflaint]	20.43		
Integral tooth flank temperature (°C)	[θint]	124.94		

Required safety	[SSmin]	1.800
Safety factor for scuffing (intg.-temp.)	[SSint]	3.643
Safety factor for transmitted moment (int.-T.)	[SSL]	8.348

12 Measurements for tooth thickness

12.1 Tooth thickness tolerances

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness tolerance		DIN 3967 cd25	DIN 3967 cd25
Tooth thickness allowance (normal section) (mm)	[Asn.e/i]	-0.095 / -0.145	-0.130 / -0.190

12.2 Base tangent lengths

		------- Gear 1 ----------	Gear 2 -------
Number of teeth spanned	[k]	4.000	8.000
Base tangent length (no backlash) (mm)	[Wk]	65.680	137.540
Base tangent length with allowance (mm)	[Wk.e/i]	65.591 /65.544	137.418 /137.362
(mm)	[ΔWk.e/i]	-0.089 / -0.136	-0.122 / -0.179
Diameter of measuring circle (mm)	[dMWk.m]	145.311	407.268

12.3 Measurement over balls and pins

		------- Gear 1 ----------	Gear 2 -------
Theoretical diameter of ball/pin (mm)	[DM]	11.448	9.996
Effective diameter of ball/pin (mm)	[DMeff]	12.000	10.000
Radial single-ball measurement, no backlash (mm)	[MrK]	80.727	209.180
Radial single-ball measurement (mm)	[MrK.e/i]	80.638 /80.590	209.002 /208.920
Diameter of measuring circle (mm)	[dMMr.m]	143.669	404.616
Diametral measurement over two balls, no backlash (mm)	[MdK]	161.107	418.360
Diametral measurement over two balls (mm)	[MdK.e/i]	160.927 /160.832	418.005 /417.839
Diametral measurement over pins, no backlash (mm)	[MdR]	161.107	418.360
Measurement over pins according to DIN 3960 (mm)	[MdR.e/i]	160.927 /160.832	418.005 /417.839
Measurement over 3 pins, no backlash (mm)	[Md3R]	160.759	0.000
Measurement over 3 pins (mm)	[Md3R.e/i]	160.580 /160.485	0.000 / 0.000

12.4 Tooth thickness

		------- Gear 1 ----------	Gear 2 -------
Medium tip diameter (mm)	[da.m]	155.044	416.938
Reference chordal height from da.m (mm)	[hac]	8.753	4.512
Tooth thickness at height hac, chord (mm)	[sc]	11.277	8.339
(mm)	[sc.e/i]	11.185 /11.136	8.210 / 8.150
Tooth thickness on reference circle, arc (mm)	[sn]	11.290	8.339
(mm)	[sn.e/i]	11.195 /11.145	8.209 / 8.149

12.5 Backlash

		---------- Pair ----------
Circumferential backlash from Aa (mm)	[jtw_Aa.e/i]	0.012 / -0.012
Radial backlash (mm)	[jrw.e/i]	0.466 / 0.286
Circumferential backlash (transverse section) (mm)	[jtw.e/i]	0.348 / 0.214
Normal backlash (mm)	[jnw.e/i]	0.326 / 0.200
Center distance allowances (mm)	[Aa.e/i]	0.016 / -0.016
Backlash free center distance (mm)	[aControl.e/i]	273.754 /273.606

Backlash free center distance, allowances (mm)	[jta]	-0.302 / -0.450	

		------- Gear 1 ----------	Gear 2 -------
dNf.i with aControl (mm)	[dNf0.i]	132.314	397.560
Reserve (dNf0.i-dFf.e)/2 (mm)	[cF0.i]	-0.005	1.322
Tip clearance (mm)	[c0.i(aControl)]	1.228	1.180

Torsional angle Gear 2 with fixed Gear 1:			
Total torsional angle (°)	[j.tSys]	0.0975/ 0.0598	(0°5'51"/ 0°3'35")
Torsional angle Gear 1 with fixed Gear 2:			
Total torsional angle (°)	[j.tSys]	0.2881/ 0.1769	(0°17'17"/ 0°10'37")

13 Gear tolerances

		------- Gear 1 ----------	Gear 2 -------
According to ISO 1328-1:2013, ISO 1328-2:1997			
Accuracy grade	[Q]	A6	A6
Single pitch tolerance (µm)	[fpT]	±11.00	±11.00
Base pitch tolerance (µm)	[fpbT]	±10.34	±10.34
Sector pitch tolerance (µm)	[Fpz/8T]	±22.00	±26.00
Cumulative pitch tolerance, total (µm)	[FpT]	32.00	40.00
Adjacent pitch difference tolerance (µm)	[fuT]	15.00	16.00
Profile form tolerance (µm)	[ffαT]	12.00	12.00
Profile slope tolerance (µm)	[fHαT]	±9.00	±9.50
Profile tolerance, total (µm)	[FαT]	15.00	15.00
Helix form tolerance (µm)	[ffβT]	10.00	11.00
Helix slope tolerance (µm)	[fHβT]	±9.00	±9.50
Helix tolerance, total (µm)	[FβT]	14.00	15.00
Runout tolerance (µm)	[FrT]	29.00	36.00
Single flank composite tolerance, total (µm)	[FisT]	43.00	50.00
Single flank composite tolerance, tooth-to-tooth (µm)	[fisT]	10.00	10.00
Radial composite tolerance, total (µm)	[Fi"]	51.00	60.00
Radial composite tolerance, tooth-to-tooth (µm)	[fi"]	22.00	22.00

F_{idT} (F_i"), f_{idT} (f_i") according to ISO 1328-2:1997 calculated with the geometric mean values for m_n and d.

According to ISO 1328-2:2020			
Accuracy grade	[R]	R41	R41
Radial composite tolerance, total (µm)	[FidT]	45.00	57.00
Radial composite tolerance, tooth-to-tooth (µm)	[fidT]	21.00	24.00

Axis alignment tolerances recommendation according to ISO/TR 10064-3:1996, AGMA 915-3-A99

Total helix deviation for calculation (µm)	[Fβ]	15.00
Distance between bearings for calculation (mm)	[LG]	56.60
Maximum value for axis deviation error (µm)	[fΣβ]	± 15.00
Maximum value for axis inclination error (µm)	[fΣδ]	± 30.00

14 Modifications and determination of the tooth form

14.1 Data for the tooth form calculation

Data not available.
Please run the calculation in the "Tooth form" tab and open the main report again.

15 Supplementary data

Maximal possible centre distance (ε_α = 1.0)	[aMAX]	277.884

15.1 Masses, stiffnesses and moments of inertia

		------- Gear 1	---------- Gear 2 -------
Mass (kg)	[m]	2.692	9.441
Total mass (kg)	[Σm]	12.133	
Moment of inertia for system, relative to the input:			
calculation without consideration of the exact tooth shape			
Gears individually ((da+df)/2...di) (kg*m²)	[J]	0.00829	0.32026
System (da+df)/2...di (kg*m²)	[J]	0.04493	

15.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:		
Torsional stiffness (MNm/rad)	[cr]	1.836
Torsion when subjected to nominal torque (°)	[δcr]	0.051
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	78.285

15.3 Power loss

Average coefficient of friction according to Niemann	[μ_m]	0.069
Wear sliding coefficient by Niemann	[ζw]	0.856
Loss factor	[HV]	0.143
Gear power loss (kW)	[PVZ]	0.736
Meshing efficiency (%)	[ηz]	99.018

16 Service life, damage

Required safety for tooth root	[S_{Fmin}]	1.4000
Required safety for tooth flank	[S_{Hmin}]	1.0000
Required service life	[H]	20000.0000

Service life (calculated with required safeties):

System service life (h)	[H_{att}]	16454

		------- Gear 1	---------- Gear 2 -------
Tooth root service life (h)	[H_{Fatt}]	1e+06	1.645e+04
Tooth flank service life (h)	[H_{Hatt}]	1.697e+04	5.018e+04

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

16.1 Damage

Damage relative to the required service life (H, 20000.0 h)

D_{F1} (%)	<TD CENTER>D_{F2} (%)	<TD CENTER>D_{H1} (%)	<TD CENTER>D_{H2} (%)
0.0000	121.5484	117.8347	39.8559

Damage relative to the system service life (H_{att}, 16454.3 h)

D_{F1} (%)	<TD CENTER>D_{F2} (%)	<TD CENTER>D_{H1} (%)	<TD CENTER>D_{H2} (%)
0.0000	100.0000	96.9447	32.7901

17 Reliability calculation

17.1 Calculation method

Calculation method according to B. Bertsche, Reliability in Automotive and Mechanical Engineering, Springer-Verlag Berlin Heidelberg 2008

17.2 Factors

Reliability of material data for σ_{Hlim} (%)	[$R\sigma_{Hlim}$]	99.00

Reliability of material data for σ_{Flim} (%) [$R\sigma_{Flim}$] 99.00

Calculation of coefficients for reliability R(t)
$R(t) = 100 \cdot Exp(-((t \cdot fac - t_0)/(T - t_0))^{\beta})$ (%)

Gear	Type	fac	<TD CENTER>β	<TD CENTER>t_0	<TD CENTER>T	<TD CENTER>R(H)
		cycles/h		cycles	cycles	%
1	Tooth root	26448	1.700e+00	9.654e+29	1.484e+30	100.0000
1	Tooth flank	26448	1.300e+00	4.046e+08	1.928e+09	96.2246
2	Tooth root	8946	1.700e+00	1.421e+08	2.184e+08	74.8383
2	Tooth flank	8946	1.300e+00	4.046e+08	1.928e+09	100.0000

fac = Number of load cycles per hour
β = Weibull shape parameter
t_0 = Failure-free number of load cycles
T = Characteristic service life (in load cycles) for 63.2% failure probability
R(H) = Reliability for required service life

17.3 Resulting reliabilities and service lifes

Required service life (h)	[H_{min}]	20000.0000
Reliability R, tooth roots subsystem (%)	[R_{subF}]	74.8383
Reliability R, tooth flanks subsystem (%)	[R_{subH}]	96.2246
Reliability R, gears subsystem (%)	[R_{subG}]	72.0129
Required reliability (%)	[R_{min}]	99.0000
Service life H, tooth roots subsystem (h)	[H_{subF}]	16454.3464
Service life H, tooth flanks subsystem (h)	[H_{subH}]	16972.9255
Service life H, gears subsystem (h)	[H_{subG}]	16267.6407

18 Remarks

18.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.

- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.

18.2 Calculations and factors

- Details of calculation method:
  c_γ according to Method B
  K_v according to Method B
  $K_{H\beta}$ and $K_{F\beta}$ according to Method C
  $f_{m\alpha}$ according to equation 66, f_{sh} according to 59/60, $F_{\beta x}$ according to 54/55/57
  $K_{H\alpha}$, $K_{F\alpha}$ according to Method B
- The logarithmically interpolated value must be entered for factors Z_L, Z_V, Z_R, Z_W, Z_X, $Y_{\delta relT}$, Y_{RrelT} and Y_X. This value is calculated from the infinite life strength and static strength values, based on the number of load cycles.

End of report (lines: 733)